

# Practical 3

## Time Series Features: Estimation and forecasting

### Getting started

At the end of this practical you should be able to:

- Manipulate data with transformations, differencing, and seasonal differencing.
- Investigate residuals, identify further structure and check if they are stationary.
- Produce and evaluate forecasts for models.

Before getting started, you may find it beneficial to read these short introductions to some of the packages we will use:

- [Introduction to fable](#)
  - **fable** 0.4.1 (O'Hara-Wild, Hyndman, and Wang 2024)

### The Mauna Loa Data

Recall for the Mauna Loa Data, the residual series after fitting a cubic regression with seasonal effects was obtained as follows.

```
library(tsibble)
library(dplyr)
library(ggplot2)
library(fable)

mauna <- as_tsibble(co2) |>
  # use meaningful names
  rename(year_month = index, concentration = value)

mauna_model_cs <- mauna |>
  model(
    lm_cs = TSLM(concentration ~ trend() + I(trend()^2) + I(trend()^3) + season()),
  )

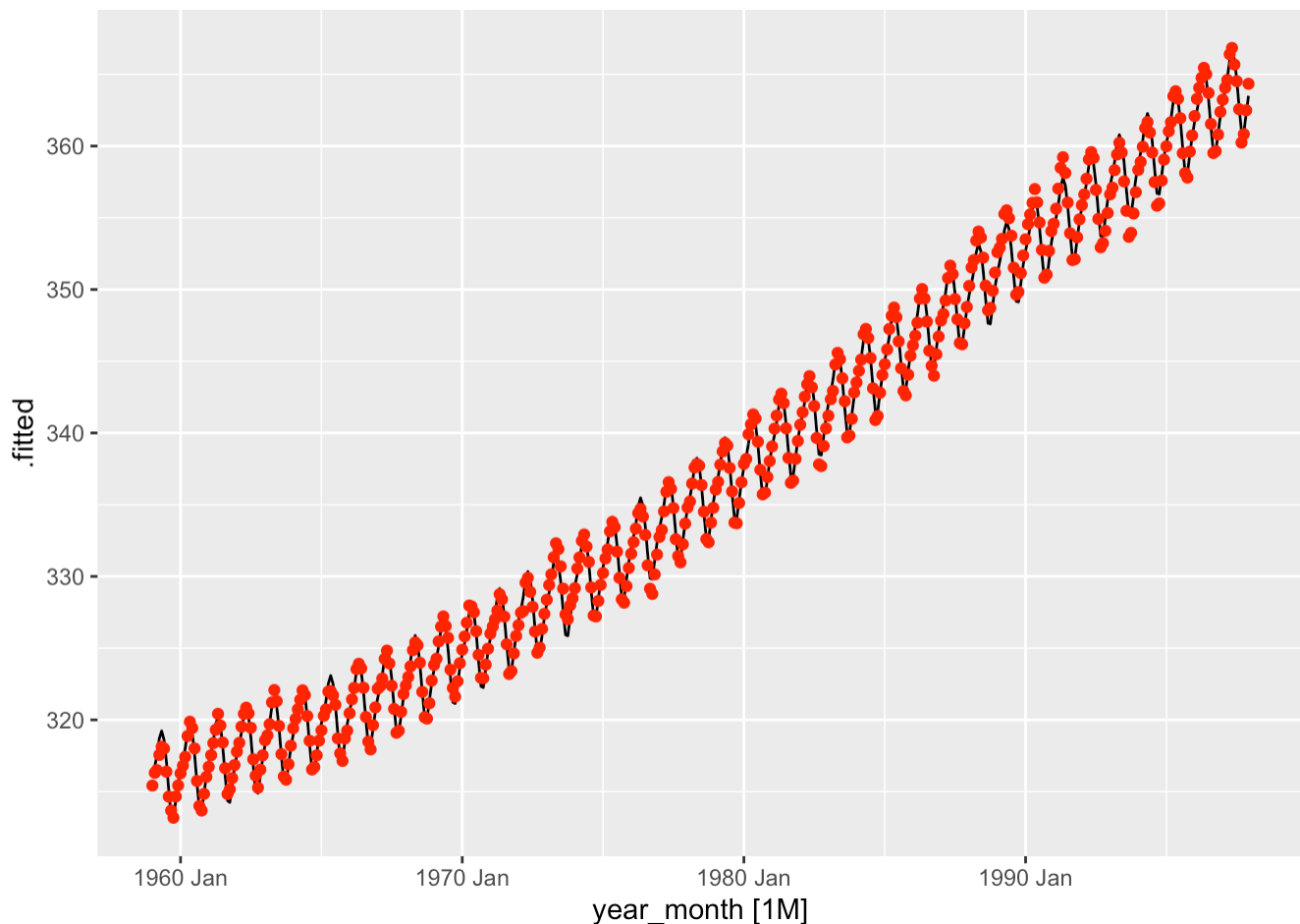
tidy(mauna_model_cs)
```

# A tibble: 15 × 6

|   | .model | term         | estimate     | std.error    | statistic | p.value   |
|---|--------|--------------|--------------|--------------|-----------|-----------|
|   | <chr>  | <chr>        | <dbl>        | <dbl>        | <dbl>     | <dbl>     |
| 1 | lm_cs  | (Intercept)  | 316.         | 0.121        | 2612.     | 0         |
| 2 | lm_cs  | trend()      | 0.0328       | 0.00174      | 18.8      | 7.88e- 59 |
| 3 | lm_cs  | I(trend()^2) | 0.000274     | 0.00000861   | 31.9      | 1.08e-117 |
| 4 | lm_cs  | I(trend()^3) | -0.000000264 | 0.0000000121 | -21.9     | 7.20e- 73 |

|    |       |                |        |       |       |           |
|----|-------|----------------|--------|-------|-------|-----------|
| 5  | lm_cs | season()year2  | 0.670  | 0.114 | 5.85  | 9.32e- 9  |
| 6  | lm_cs | season()year3  | 1.42   | 0.114 | 12.4  | 1.45e- 30 |
| 7  | lm_cs | season()year4  | 2.56   | 0.114 | 22.3  | 5.61e- 75 |
| 8  | lm_cs | season()year5  | 3.04   | 0.115 | 26.6  | 2.31e- 94 |
| 9  | lm_cs | season()year6  | 2.38   | 0.115 | 20.8  | 5.37e- 68 |
| 10 | lm_cs | season()year7  | 0.868  | 0.115 | 7.58  | 2.00e- 13 |
| 11 | lm_cs | season()year8  | -1.19  | 0.115 | -10.4 | 5.72e- 23 |
| 12 | lm_cs | season()year9  | -3.01  | 0.115 | -26.3 | 2.79e- 93 |
| 13 | lm_cs | season()year10 | -3.19  | 0.115 | -27.9 | 3.04e-100 |
| 14 | lm_cs | season()year11 | -2.00  | 0.115 | -17.4 | 1.97e- 52 |
| 15 | lm_cs | season()year12 | -0.874 | 0.115 | -7.63 | 1.41e- 13 |

```
augment(mauna_model_cs) |> autoplot(.fitted) +  
  geom_point(aes(x=year_month,y=concentration), colour = "red") # add raw data
```



1. Obtain plots of the model residuals: the ACF and the PACF.
2. Do the residual series appear stationary? Use plots and the portmanteau test.
3. Do you think this model is a good fit for the data? Why or why not? What evidence will you use to support your conclusion?

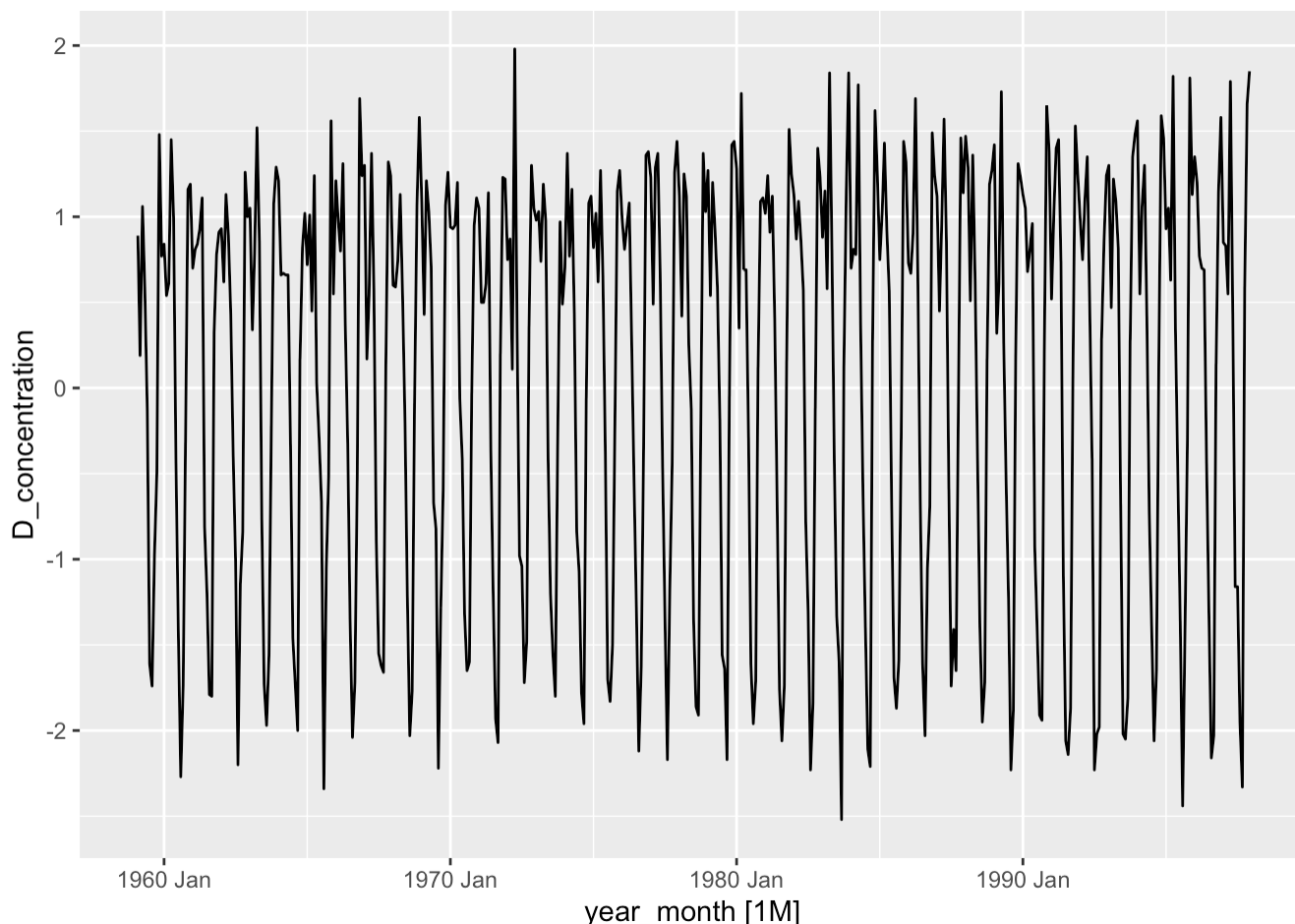
## Evaluating differences

## Exploring differencing

Differencing can be achieved using the `difference` function. For example, if  $X_t$  is concentration then  $\nabla X_t$  can be calculated with the following code.

```
mauna <- mauna |> mutate(D_concentration = difference(concentration, lag = 1))

mauna |> autoplot(D_concentration)
```



Recall  $\nabla = 1 - B$ , in contrast seasonal differencing can be defined as  $\nabla_s = 1 - B^s$ , which can be calculated using `difference(series, lag = s)`.

4. Apply seasonal differencing to `concentration` and `D_concentration`, investigate the residuals, does either transformed series appear stationary?
5. Apply seasonal then standard differencing to create a new variable called `DS_concentration`, does this transformed series appear stationary? Does it matter if the seasonal differencing is performed first?

## AR model

Since we have identified further structure in the data set, we should try to model this. We can fit an

Since we have identified further structure in the data set, we should try to model this. We can fit an AR model to the differenced data as follows.

```
mauna_model_ar <- mauna |>
  # remove NAs from differencing before model
  filter(!is.na(DS_concentration)) |>
  model(
    DS_ar1 = AR(DS_concentration ~ order(1))
  )

tidy(mauna_model_ar)
```

```
# A tibble: 1 × 6
  .model term estimate std.error statistic p.value
  <chr>   <chr>      <dbl>      <dbl>      <dbl>   <dbl>
1 DS_ar1 ar1      -0.302      0.0447      -6.76 4.27e-11
```

- Investigate the model residuals, do they appear stationary? Is there any additional correlation structure that needs to be modeled?

## Differencing in the model

ARIMA models (which we will cover thoroughly in the next module) can support differencing by construction<sup>1</sup>. The ARIMA model below replicates the differencing and AR(1) structure we produced manually earlier.

```
mauna_model_arima <- mauna |>
  model(
    arima1 = ARIMA(concentration ~ pdq(1,1,0) + PDQ(0,1,0))
  )
```

- What does `pdq` specify in the model? How about `PDQ`? Are the residuals of this model related to the residuals of the `DS_ar1` model?
- Investigate the model residuals, do they appear stationary? Is there any additional correlation structure that needs to be modeled?

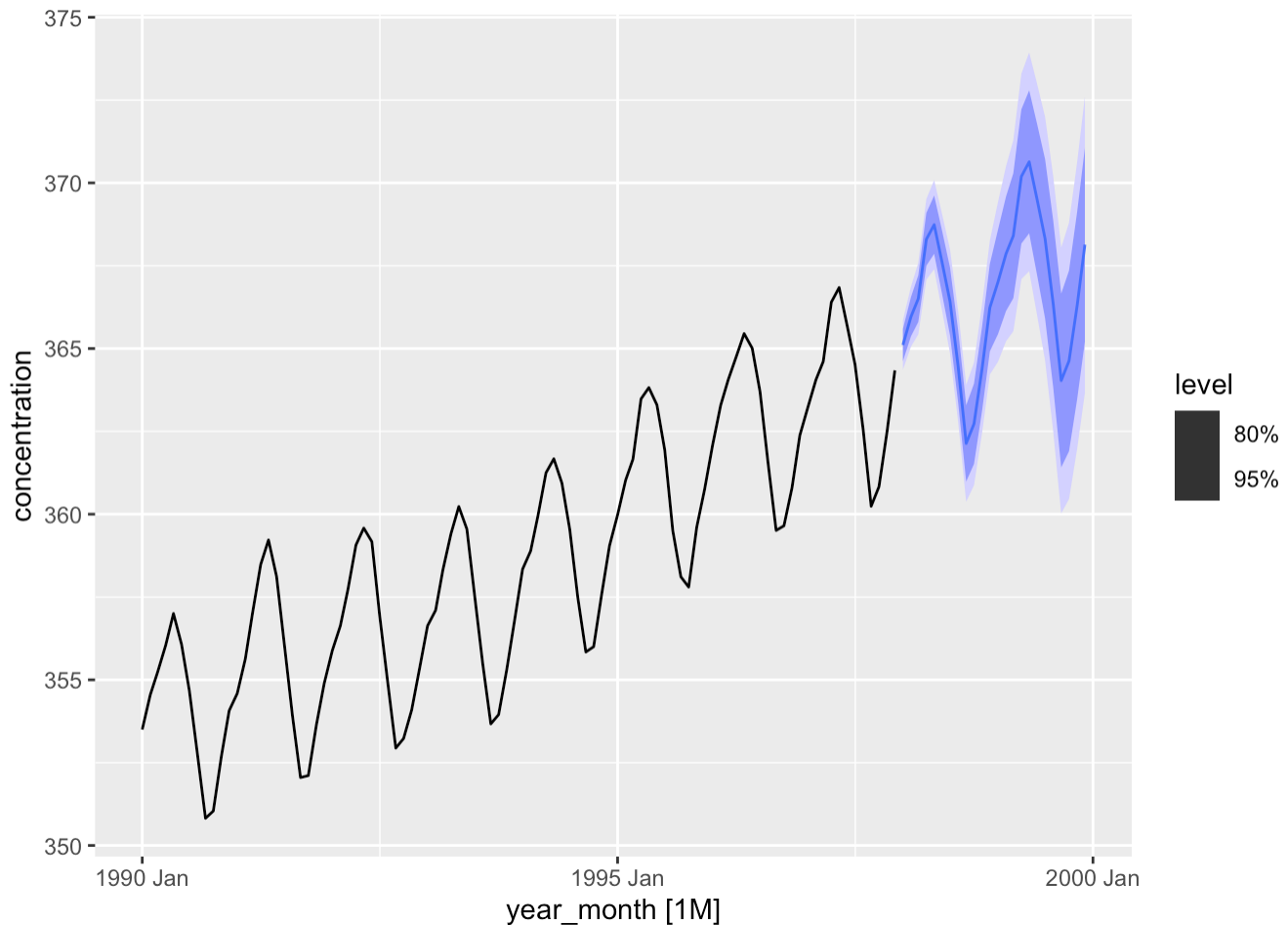
## Forecasting

The `forecast` function will produce a probabilistic forecast for models from the `fable` package. This can be done as follows.

```
arima_fc <- mauna_model_arima |> forecast()

mauna |>
  # filter to data after Jan 1990
```

```
filter(year_month >= yearmonth("Jan 1990")) |>
autoplot(concentration) +
autolayer(arima_fc)
```



9. Refit the `mauna_model_arima` model (or your chosen model) with a subset of the data, say Jan 1959 to Dec 1994, and produce a forecast from Jan 1995 to Dec 1997.
10. Plot the forecast and actual data observed. Is this a good forecast of the data? How could you evaluate this empirically?

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## References

O'Hara-Wild, Mitchell, Rob Hyndman, and Earo Wang. 2024. *Fable: Forecasting Models for Tidy Time Series*. <https://doi.org/10.32614/CRAN.package.fable>.

## Footnotes

1. As such ARIMA models can handle some types of non-stationarity. [↗](#)